

Máximo Fernández Núñez

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AI engineer, +13 years shipping production systems. Focused on LLM agents: architecture, tooling, observability and deployment. Background in computer vision and embedded AI on edge hardware.

PROFESSIONAL EXPERIENCE

Unusuals

AI Agent Engineer

Remote

January 2026 – July 2026

- Sole engineer behind the agent of an inspection assistant for Spain's national grid operator (REE), letting field technicians query millions of high-voltage pylon photos by region and defect type instead of reviewing towers one by one. In production, used by real teams.
- Redesigned the architecture from single intent-to-tool routing to a multi-step ReAct loop over ~10 tools (image analysis, S3 navigation, vision-model training, email reporting), turning a command-driven bot into an autonomous agent able to resolve multi-region requests end to end in one conversation.
- Added multi-provider LLM routing (OpenAI, Mistral, OpenRouter) with automatic failover on provider outages, plus step limits, guardrails and prompt caching to reduce token spend.
- Introduced LangSmith tracing: root-cause analysis dropped from 10+ minutes of manual log digging to seconds, and user-reported failures became exactly reproducible, previously they took days or weeks to reproduce.
- Built per-user memory that adapts to individual preferences, slash commands, and Microsoft Teams and web interfaces. Owned deployment and infrastructure.

Freelance

Generative AI | Agents | RAG | chatbots

Remote

December 2024 – January 2026

- Delivered a multi-agent tutoring system (one agent per subject) for Turing Dream, used by secondary-school students in Spain and Colombia, with course material authored and validated by a team of teachers. Agent planning, sub-agents, tool design, context engineering and human-in-the-loop.
- Built a voice agent for kitchen and floor staff at CIVIR to consult and prepare recipes and menus. Designed a streaming pipeline (100 ms audio chunks, VAD-gated buffering, silence-based end-of-speech detection) that sent only real speech to the STT API, sharply cutting transcription cost.
- Owned backend, frontend and deployment of the agent side of the product, working alongside a 3-person team on the SaaS platform and infrastructure. Direct requirements gathering with clients.

Bravent

AI Tech Lead

Remote

Sept 2024 – November 2024

- Technical leadership on a generative-AI initiative on Azure: architecture, technology selection and risk assessment. The programme was halted by the client.

Sener Aeroespacial

Machine learning engineer

Madrid, Spain

Sept 2022 – Sept 2024

- Built the perception stack (segmentation, classification, detection, depth estimation) for an autonomous ground vehicle operating in unstructured terrain, within a European defence consortium.
- Optimised inference with TensorRT on NVIDIA AGX Orin: ~30% lower latency and VRAM usage for ~2% accuracy loss, fitting the full stack on embedded hardware.
- Created the segmentation dataset from scratch and fine-tuned models where no public dataset covered the target objects. Built an annotation tool that cut labelling time per image from hours to minutes.
- Supervised a colleague's master's thesis on a RAG system over internal project documentation. Mentored 3 interns and junior engineers, and successfully backed a vision-team colleague's case for promotion.

Arquimea Machine Learning & Embedded Engineer	Madrid, Spain <i>October 2017 – July 2022</i>
- Proposed and delivered a single hardware platform with shared low-level firmware across 3 UAV models, eliminating per-model HW/FW redesign and letting features built for one aircraft be inherited by the rest. - Replaced a commercial object-detection module with an in-house YOLO detector, saving €2,000 per unit. - Designed schematics and PCBs and programmed the microcontrollers for UAV flight control. One resulting system is deployed at several airports for runway bird dispersal. - Prototyped GPS-denied geolocation: trained a network on overhead imagery of Spain and France that localised a UAV from a single nadir image to ~1 km error, enough to return an aircraft to friendly territory when GPS is jammed, instead of losing it behind enemy lines. MVP, not taken to production.	
Indra Electronic engineer	Madrid, Spain <i>January 2013 – Oct 2017</i>
Designed control PCBs and firmware for an autonomous helicopter programme for the Spanish Ministry of Defence.	
EDUCATION	
Universidad Complutense de Madrid Electronic engineering	Madrid, Spain <i>June 2011</i>
Honors in the subjects of Systems control, Robotics, Introduction to Semiconductors, Computer architecture and in Final degree project.	
PROJECTS & COMMUNITY	
Gymnasia (github.com/maximofn/gymnasia): Fitness app running two agents fully on-device, with no backend: user data never leaves the phone. A BYOK conversational coach (adapters for OpenAI, Anthropic and Google, 12 local tools, streaming tool-use loop, remote system prompt with offline fallback) and a vision sub-agent that estimates macros from meal photos, with barcode lookup against OpenFoodFacts. Architecture documented at gymnasia-sable.vercel.app. Welcome to la secta (youtube.com/@Welcometolasecta): Weekly AI podcast, 800 subscribers, ~500 views per episode, with guests from the Spanish-speaking AI scene. Built a generative-AI pipeline that renders, uploads, clips and distributes each episode across platforms automatically, including voice cloning and lip-sync for future multilingual editions. maximofn.com: AI blog and technical writing, built and self-operated: Astro on Vercel behind Cloudflare as reverse proxy, WAF and media CDN. ~2,000 LinkedIn followers, posting near-daily.	
SKILLS	
Agents & LLMs: LangGraph, LangChain, LangSmith, MCP, OpenAI Agents SDK, ReAct architectures, RAG, vector databases, guardrails, prompt caching, multi-provider failover ML & vision: PyTorch, TensorRT, HuggingFace, YOLO, segmentation, detection, depth estimation, fine-tuning, dataset creation Engineering: Python, SQL, Docker, git, CI/CD, testing, code review, bash/SSH, Gradio, Astro, HTML/CSS Cloud & edge: Azure (ML, Functions), AWS S3, Railway, Vercel, Cloudflare (reverse proxy, WAF, DDoS mitigation, CDN); NVIDIA Jetson / AGX Orin, embedded firmware, PCB design Languages: Spanish (native), English (full professional proficiency), Portuguese (elementary)	